

Employee Attrition Prediction Using Machine Learning

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Abstract—This paper presents an employee attrition prediction system built using a stacked ensemble classifier combining Random Forest and XGBoost as base learners with Logistic Regression as a meta-learner. Applied to a combined IBM HR dataset of 11,058 records, the model achieves 98.23% accuracy, 99.65% precision, 96.79% recall, 98.20% F1-score, and 98.74% AUC-ROC — outperforming all eight evaluated individual classifiers and four prior state-of-the-art models on the same benchmark dataset. PCA-based dimensionality reduction and class-weight balancing further enhance model robustness. The proposed system provides HR professionals with a reliable, data-driven tool for identifying at-risk employees and designing targeted retention strategies.

Index Terms—Employee Attrition, Stacked Ensemble Learning, Random Forest, XGBoost, Logistic Regression, PCA, Class Imbalance, HR Analytics, Machine Learning.

I. PROBLEM STATEMENT

Employee attrition — the voluntary or involuntary departure of employees from an organization — has emerged as a critical challenge in the post-pandemic era. The phenomenon, widely termed the 'Great Resignation,' imposes significant operational and financial burdens across all sectors, most acutely in the Information Technology (IT) industry [1].

High attrition rates impose both direct and indirect costs on organizations:

- Direct costs: recruitment expenses, onboarding, training, and reduced productivity during transition periods.
- Indirect costs: loss of implicit knowledge, erosion of team morale, damage to organizational reputation, and disruption of workflow continuity.

The key issues driving this problem are:

- HR managers lack data-driven tools to identify at-risk employees in time for preventive intervention.
- Traditional ML classifiers for attrition prediction operate as black boxes, offering predictions without interpretable reasoning.
- Imbalanced class distributions in HR datasets skew model predictions toward the majority (non-attrition) class.
- Most existing models are trained on single, limited datasets, reducing their generalizability.

This project addresses these gaps by building a high-accuracy attrition prediction system using a stacked ensemble model applied to a combined and extended IBM HR dataset with 11,058 employee records.

II. OBJECTIVE

The primary objective is to develop a robust and interpretable employee attrition prediction system using advanced machine learning techniques. Specific objectives are:

- 1) Develop a stacked ensemble classifier using Random Forest (RF) and XGBoost as base classifiers with Logistic Regression as the meta-learner.
- 2) Combine and preprocess two HR datasets to produce a comprehensive dataset of 11,058 employee records with relevant features.
- 3) Address class imbalance using class-weight balancing and the `scale_pos_weight` parameter in XGBoost, without distorting the original data distribution.
- 4) Apply Principal Component Analysis (PCA) for dimensionality reduction, retaining the most significant variance-preserving components.
- 5) Evaluate and compare eight ML classifiers — LR, RF, XGBoost, AdaBoost, SVM, KNN, LightGBM, and DT — to select the best-performing base models for stacking.
- 6) Achieve high predictive performance targeting accuracy and F1-score exceeding state-of-the-art baseline models on the same dataset.

The expected outcome is an accurate, reliable, and actionable attrition prediction system that enables HR professionals to identify at-risk employees and implement effective retention strategies.

III. EXISTING WORK

A. Literature Review

A systematic review of eleven key studies was conducted. Table I summarizes the reviewed papers along with their methodologies, datasets, and key contributions.

TABLE I: SUMMARY OF LITERATURE REVIEWED

S.No	Authors	Year	Key Contribution
1	Lazzari et al.	2022	KNN, DT, LR, XGB, LGBM; causality-based; predictive feature prioritization
2	Chung et al.	2023	Stacking (LR, RF, XGBoost, SVM, ANN); satisfaction & overtime as key factors
3	Raza et al.	2022	Optimized Extra Tree Classifier; monthly income, job level, age as factors
4	Duan et al.	2025	Deep learning CNN with image representation of tabular data; 97.8% accuracy

S.No	Authors	Year	Key Contribution
5	Al Akasheh et al.	2024	Knowledge graph + GCN; XAI for interpretability; linear SVM best performer
6	Biswas et al.	2023	Ensemble (GB, RF, KNN); professional networking as strongest predictor
7	Gim et al.	2023	RF, XGBoost, ANN on 35 variables; XGBoost best accuracy, RF best precision
8	Alsheref et al.	2022	Automated ML pipeline (RF, MLP, GB) with auto-tuning hyperparameters
9	Shafie et al.	2024	ANN with clustering and CTGAN data augmentation; addressed class imbalance
10	Park et al.	2024	LR, KNN, XGBoost; workload and job security as key influencing factors
11	Konar et al.	2025	RF+XGB+LR stacking with Bayesian optimization; SHAP+LIME; 98.8% accuracy [BASE PAPER]

B. Base Paper for Implementation

Title: Employee Attrition Prediction Using Bayesian Optimized Stacked Ensemble Learning and Explainable AI [1].

Authors: Karabi Konar, Saptarshi Das, Samiran Das, Sanjay Misra. Published in: SN Computer Science, Vol. 6, Article 672, July 2025. DOI: 10.1007/s42979-025-04204-w.

Key Methodology: Stacked ensemble combining RF and XGBoost as base learners with Logistic Regression as meta-learner. Bayesian optimization for hyperparameter tuning. SHAP and LIME for model interpretability. Trained on IBM HR Analytics dataset; achieves 98.8% accuracy and 98.5% F1-score.

Why this paper was selected:

- Addresses both prediction accuracy and model interpretability — critical challenges in HR analytics.
- The stacked ensemble framework provides a strong foundation extensible to larger datasets.
- Bayesian optimization offers an efficient alternative to traditional grid/random search.
- Methodology is well-documented and reproducible, suitable for replication and extension.

IV. RESEARCH GAP

A. Single-Classifer Dependence

Many existing studies rely on a single ML model such as SVM or Gradient Boosting. These are prone to overfitting on imbalanced data and lack the robustness that ensemble approaches provide, limiting generalizability across organizational contexts.

B. Lack of Explainability

Models such as neural networks, SVM, and gradient boosting operate as black boxes. Without insight into decision-making, HR professionals cannot trust or interpret predictions, reducing practical adoption in real HR environments.

C. Small and Limited Datasets

Most studies use the IBM HR dataset with only 1,470 records — insufficient for building robust and generalizable models. The limited sample restricts the model's ability to learn complex, cross-organizational attrition patterns.

D. Inadequate Class Imbalance Handling

HR datasets are inherently imbalanced (~4:1 ratio). Many approaches apply SMOTE which may introduce noise by synthesizing minority class samples, particularly problematic with high proportions of categorical features.

E. Missing Feature Engineering and Optimization

Several studies skip PCA-based dimensionality reduction and hyperparameter optimization, causing models to learn from noisy or redundant features and operate at suboptimal settings.

F. How This Work Addresses the Gaps

- RF + XGBoost + LR meta-learner stacking addresses single-classifier limitations by combining diverse learner strengths.
- Combined dataset of 11,058 records (7.5× larger) provides richer training and better generalization.
- PCA eliminates redundant features and reduces overfitting risk.
- Class-weight balancing addresses the 4:1 imbalance without distorting the original data distribution.

V. PROPOSED METHODOLOGY

A. System Overview

The proposed system follows a structured end-to-end ML pipeline comprising six stages: dataset collection and merging, preprocessing, class imbalance handling, dimensionality reduction via PCA, stacked model training, and performance evaluation.

Collection → Preprocessing → Balancing → PCA → Training → Evaluation

B. Data Collection

Two publicly available datasets were used:

- IBM HR Analytics Employee Attrition Dataset (Kaggle, ~1,470 records) — the benchmark dataset used in the base paper [1].
- Extended alternate HR attrition dataset from Kaggle, containing records across diverse departments and roles [2].

These datasets were merged and cleaned to produce a final combined dataset of 11,058 employee records with 22 features. Class distribution after merging: 8,882 non-attrition (0) and 2,176 attrition (1) — a class imbalance ratio of approximately 4.08:1.

TABLE II: COMBINED DATASET FEATURE DESCRIPTION

Feature	Type	Description
Age	Integer	Age of the employee
JobRole	Categorical	Job role (Manager, Assistant, etc.)
Department	Categorical	Dept. (HR, Finance, R&D, Sales, etc.)
JobLevel	Integer	Seniority level (1–5)
YearsAtCompany	Integer	Years worked at the company
MonthlyIncome	Float	Monthly salary
PercentSalaryHike	Float	Salary increase over previous year (%)
JobSatisfaction	Integer	Job satisfaction level (1–4)
EnvironmentSatisfaction	Integer	Work environment satisfaction (1–4)
WorkLifeBalance	Integer	Work-life balance rating (1–4)
JobInvolvement	Integer	Level of job involvement (1–4)
OverTime	Categorical	Whether employee works overtime (Yes/No)
DistanceFromHome	Integer	Distance from home to office (km)
MaritalStatus	Categorical	Marital status (Single/Married/Divorced)
NumCompaniesWorked	Integer	Number of previous companies worked
TrainingTimesLastYear	Integer	Training sessions attended last year
Attrition	Binary	Target variable: left (1=Yes, 0=No)

C. Data Preprocessing

1) Irrelevant Feature Removal: Uniform-value columns (e.g., EmployeeCount, StandardHours, Over18) were removed. Only features with discriminative value for attrition prediction were retained.

2) Label Encoding and One-Hot Encoding: The binary target 'Attrition' was label-encoded (Yes=1, No=0). Categorical features — JobRole, Gender, Department, MaritalStatus, OverTime — were one-hot encoded, expanding the feature space from 21 to 43 dimensions.

3) Min-Max Normalization: All numerical features were scaled to the [0, 1] range using MinMaxScaler to prevent high-magnitude features (e.g., MonthlyIncome) from dominating learning algorithms.

4) Class Imbalance Handling: class_weight='balanced' was applied to classifiers that support it; scale_pos_weight was set proportionally for XGBoost. This approach was preferred over SMOTE given the high proportion of categorical features in the dataset.

5) Dimensionality Reduction (PCA): PCA was applied after encoding and scaling, reducing 42 features to 24 principal components. Significant contributing original features included: EnvironmentSatisfaction, JobInvolvement, WorkLifeBalance, MonthlyIncome, OverTime, DistanceFromHome, and JobSatisfaction.

D. Model / Algorithms Used

Eight base classifiers were trained and evaluated to select best-performing models for the stacking architecture:

- Logistic Regression (LR) — baseline linear model.
- Random Forest (RF) — bagging ensemble of decision trees.
- XGBoost — gradient boosting with regularization.
- AdaBoost — adaptive boosting.
- Support Vector Machine (SVM) with RBF kernel.
- K-Nearest Neighbor (KNN).
- LightGBM — leaf-wise gradient boosting.
- Decision Tree (DT) — CART algorithm.

Based on performance evaluation, RF and XGBoost were selected as base learners for the Stacking Classifier, with Logistic Regression serving as the meta-learner.

E. Implementation Steps

- 7) Installed required libraries: scikit-learn, XGBoost, LightGBM, scikit-optimize, imbalanced-learn.
- 8) Loaded and merged two HR datasets into a combined dataframe of 11,058 records with 22 columns.
- 9) Applied preprocessing: dropped irrelevant columns, label-encoded Attrition, one-hot encoded categoricals, normalized numericals with MinMaxScaler.
- 10) Performed stratified train-test split: 80% training (8,844 samples), 20% testing (2,214 samples).
- 11) Trained all 8 base classifiers with class_weight='balanced' or scale_pos_weight; evaluated using 5-fold cross-validation and held-out test metrics.
- 12) Applied PCA (n_components=24) on training data; transformed training and test sets accordingly.
- 13) Built Stacking Classifier: RF + XGBoost as base estimators, Logistic Regression as meta-learner, with cv=5.
- 14) Evaluated stacking model on test set: Accuracy, Precision, Recall, F1-Score, and AUC-ROC.

Tools: Python 3.x, NumPy 1.26.4, Pandas 2.3.3, Scikit-learn, XGBoost, LightGBM, scikit-optimize (BayesSearchCV), Matplotlib, Seaborn. All experiments run on Kaggle Notebooks.

VI. RESULTS AND DISCUSSION

A. Base Model Comparison

All eight classifiers were trained on the PCA-transformed combined dataset (24 components) and evaluated on the 20% held-out test split. Results are presented in Table III.

TABLE III: BASE CLASSIFIER PERFORMANCE COMPARISON

Model	Acc.%	Prec.%	Recall%	F1%	AUC%
Random Forest	91.81	92.16	91.39	91.77	96.63
XGBoost	78.92	70.93	97.97	82.29	96.22
LightGBM	82.38	79.79	86.71	83.11	90.11
KNN	76.72	68.81	97.75	80.76	95.96
SVM (RBF)	67.75	66.10	72.80	69.29	75.12
AdaBoost	50.58	50.28	99.61	66.83	50.59
Decision Tree	59.81	57.75	73.03	64.50	63.82
Logistic Reg.	54.24	54.13	55.35	54.73	56.46

Random Forest emerged as the strongest individual classifier with 91.81% accuracy and 91.77% F1-score, followed by LightGBM (83.11% F1) and XGBoost (82.29% F1). Logistic Regression and Decision Tree performed poorly due to their inability to capture non-linear feature interactions. These results confirm the selection of RF and XGBoost as base learners for the stacking architecture.

B. Stacking Classifier Results

The Stacking Classifier (RF + XGBoost base learners, Logistic Regression meta-learner) was evaluated on the 20% held-out test set. Results are presented in Table IV.

TABLE IV: STACKING CLASSIFIER PERFORMANCE

Metric	Value (%)	Interpretation
Accuracy	98.23	98.23% of all test samples correctly classified
Precision	99.65	99.65% of predicted attrition cases were true positives
Recall	96.79	Detected 96.79% of all actual attrition cases
F1-Score	98.20	Harmonic mean of precision and recall
AUC-ROC	98.74	Near-perfect class separability between attrition classes

The stacking model significantly outperforms each individual base classifier. The AUC-ROC of 98.74% confirms near-perfect class separability. The near-perfect precision of 99.65% is especially valuable in HR contexts where false positives (incorrectly flagging a loyal employee as at-risk) can damage employee trust. The combination of bagging (RF) and boosting (XGBoost) in the stacking framework effectively captures both linear and non-linear patterns in the data.

C. Comparison with Existing Studies

The proposed model was compared against four existing studies using the IBM HR Analytics dataset as a common benchmark. Results are presented in Table V.

TABLE V: COMPARISON WITH EXISTING STUDIES

Method	Acc.%	Prec.%	Recall%	F1%
Chung et al. [3]	97.6	99.9	95.1	97.5

Method	Acc.%	Prec.%	Recall%	F1%
Raza et al. [4]	93.0	93.0	93.0	93.0
Duan et al. [5]	97.8	95.5	91.3	93.4
Akashch et al. [6]	92.5	93.0	92.0	92.0
Proposed (Ours)	98.23	99.65	96.79	98.20

The proposed stacking model achieves the highest accuracy (98.23%), highest recall (96.79%), and highest F1-score (98.20%) among all compared methods. While Chung et al. [3] reported slightly higher precision (99.9%), their recall was notably lower (95.1%), indicating more missed actual attrition cases. The proposed model achieves a superior precision-recall balance, which is critical in real-world HR applications where failing to identify an at-risk employee carries significant retention cost.

VII. CONCLUSION

This report presents an employee attrition prediction system using a stacked ensemble learning model (RF + XGBoost + LR meta-learner) applied to a combined IBM HR dataset of 11,058 employee records. The system achieves high predictive accuracy while being grounded in well-validated machine learning techniques.

A. Summary of Findings

- The stacking classifier achieved 98.23% accuracy, 99.65% precision, 96.79% recall, 98.20% F1-score, and 98.74% AUC-ROC — significantly outperforming all eight individual base classifiers.
- RF (91.81% accuracy, 96.63% AUC) and XGBoost (96.22% AUC) confirmed as the most effective base learners.
- PCA compressed 42 features into 24 principal components without significant predictive performance loss.
- Class-weight balancing successfully addressed the 4:1 class imbalance in the combined dataset.
- The combined 11,058-record dataset provided a substantially richer training environment than the base paper's 1,470-record dataset, contributing to improved model robustness.

B. Limitations

- Model evaluated only on public combined datasets; generalizability to specific private organizational data not validated.
- PCA reduces direct feature interpretability since model inputs are principal components rather than original features.
- Stacking architecture increases computational overhead compared to single classifiers.
- No real-time integration with an HR management system was implemented within this project scope.

C. Future Scope

- Integration with live HR management systems for real-time attrition risk scoring and alerting.

- Extension to LSTM or Transformer architectures for temporal attrition pattern modeling.
- Validation on private, organization-specific datasets for domain generalizability assessment.
- Incorporation of SHAP/LIME explainability for transparent feature-level interpretations in future iterations.
- Development of a web dashboard displaying per-employee risk scores and feature contributions.

VIII. REFERENCES

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IX. APPENDIX

A. Dataset Sources

IBM HR Analytics Employee Attrition Dataset:

<https://www.kaggle.com/datasets/rohitsahoo/employee>

Extended Employee Attrition Dataset:

<https://www.kaggle.com/datasets/ziya07/employee-attrition-prediction-dataset>

B. Implementation

Complete Notebook (.ipynb) and Code:

<https://drive.google.com/drive/folders/1F28q3Y7JotCoptGeTbXHVPOLgQOpW0CQ?usp=sharing>

Deployment Link (Streamlit App):

<https://employee-attrition-etmawrih2nzb4ogkwrds.streamlit.app/>